VibeScent

Multimodal Fragrance Recommender via Zero-Shot Vision and Semantic Context Retrieval

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System Architecture

01	02	03	04	05	06
DATA PREP	**INPUT LAYER**	**IMAGE + TEXT PROCESSESING**	**VIBE VECTOR EXTRACTION**	**RECOMMEND**	**WEB SCRAPE**
Collecting + Processing Data	Image + Text Input	Vision + Sentence Transformers	5-Dimensonal Classification	Weighted Sum + Rank	SerpAPI + Google Shop

TECH STACK

- Qwen3-VL-Embedding-8B
- Qwen3-VL-8B
- Qwen3-Reranker-4B
- CLIP ViT-L/14
- Cosine Similarity
- SerpAPI requests

Stage 01 — Data Prep

LLM VIBE EXTRACTION | Qwen3-VL-Embedding-8B | Qwen3-VL-8B

SOURCE DATASETS

- **Parfumo** — *TidyTuesday*
- **Fragrantica Clean** — *Kaggle / olgagmiufana1*
- **Rawanalqarni** — *Kaggle*
- **rdemarqui** — *Kaggle*
- **Ayush** — *Kaggle*

~36,000
FRAGRANCES

01 Text Embedding

Qwen3-VL-Embedding-8B

- Combines fragrance notes + accords
- Embeds each fragrance scent profile
- Same model as inference pipeline

1024-d
EMBEDDING DIM

02 LLM Vibe Tag

Qwen3-VL-8B

- Vibe extracted from each fragrance
- 5 new vibe columns appended
- Allows for later comparison with user query

36K
DESCRIPTIONS GENERATED

OUTPUT SCHEMA — vibescent_enriched.csv

Formality	Season	Gender	Time_of_Day	Frequency
0.0 – 1.0	Spring · Summer · Autumn · Winter	Male · Female · Unisex	Day · Night	Often · Occasionally · Rarely

Stage 02 — Input Layer

MULTI-MODAL INPUT

MODALITY 01

Outfit Image

› **Input:** The user uploads an outfit they plan to wear

› **Goal:** Identify the outfit and fragrances that match

› **Pipeline:** Feeds into a vision transformer as raw pixel tensor

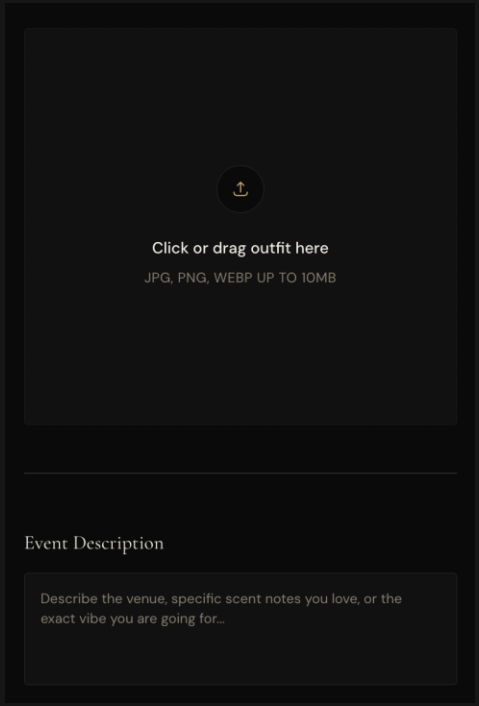
MODALITY 02

Occasion Description

› **Input:** The user describes the occasion they are dressing for

› **Goal:** Understand the occasion and choose fragrances that are appropriate

› **Pipeline:** Feeds into a sentence transformer as a string



Stage 03 — Image + Text Processing

CONTEXT | openai/clip-vit-large-patch14 | Qwen3-VL-Embedding-8B

VISION TRANSFORMER

CLIP ViT-L / 14

Vision Backbone:	Vision Transformer (ViT-L/14)
Patch Size:	14 × 14 pixels
Image Resolution:	224 × 224 (CLIPProcessor)
Embedding Dim:	768-d L2-normalized vector
Text Encoder:	Transformer (77 token context)
Contrastive Loss:	Softmax cross-entropy (ITAL)
Inference Mode:	@torch.no_grad() · .eval()
Device Target:	MPS → CUDA → CPU (auto)

SENTENCE TRANSFORMER

Qwen3-VL-Embedding-8B

Model:	Qwen/Qwen3-VL-Embedding-8B
Architecture:	Qwen3 Vision-Language Transformer
Output Dim:	1024-d vector
Max Tokens:	32,768
Pooling:	Last-token pooling
Training Obj.:	Contrastive pre-training
Warm Boot:	Loaded once at server startup
Encoding Scope:	User description

Models loaded once at FastAPI startup to eliminate cold-start latency on user query

Stage 04 — Vibe Vector Extraction

5-DIM VIBE REPRESENTATION

FORMALITY

Formality

Classes: casual · smart casual · formal

Output: `0.8 (float)`

SEASON

Season

Classes: spring · summer · fall · winter

Output: `"Winter"`

GENDER

Gender

Classes: male · female · neutral

Output: `"Male"`

TIME OF DAY

Time of Day

Classes: day · night

Output: `"Day"`

FREQUENCY

Frequency

Classes: occasional · everyday

Output: `"Often"`

Stage 05 — Recommend

MULTIMODAL LATE FUSION

Qwen3-Reranker-4B

Text

- Qwen3 embeds context input
- Text embeddings vs corpus
- Cosine similarity outputs a scalar 0-1

30%
TEXT WEIGHTING

Image

- CLIP ViT-L / 14 embeds image input
- Image embeddings vs corpus
- Cosine similarity outputs a scalar 0-1

30%
IMAGE WEIGHTING

Multimodal

- Qwen3 embeds both inputs
- Combined embedding vs corpus
- Cosine similarity outputs a scalar 0-1

25%
MULTIMODAL WEIGHTING

Structured

- Harsh fill
- Harsh fill
- Harsh fill

15%
STRUCTURED WEIGHTING

Fused Score = 0.30 (Text) + 0.30 (Image) + 0.25 (Multimodal) + 0.15 (Structured)

Reranking

- Qwen3-Reranker takes the top 20 fragrances and runs a mini-pipeline for each candidate
- Compares the input image and query with each fragrance candidate and outputs the top 3 final recommendations

Stage 06 — Web Scrape

SERPAPI SCRAPER | Amazon API

01

Amazon Shopping AP

- Engine: google_shopping via SerpAPI
- Query: '{name} perfume fragrance'
- Num results: 20 per fragrance
- Budget cap applied post-fetch
- timeout=15s · raise_for_status()

02

Priority Store Ranking

- Sephora → Nordstrom → Macy's
- Bloomingdale's → Neiman Marcus
- Ulta → Generic stores (fallback)
- _store_priority() rank function
- sort() on (priority, price) tuple

03

Result Sanitization

- None guard → fallback dict injected
- Price regex: r'[\d,]+\.?\d*'
- Float casting with ValueError guard
- Thumbnail URL passed to frontend
- purchaseUrl field for deep-link CTA

Output — Descriptive Recommendations

TOP-3 RESULTS

SELECTION NO. 01

Aventus

Creed

Pineapple | Bergamot | Birch | Musk | 94%

"A daytime winter aesthetic — birch smoke projects success."

$320.00 Sephora · WITHIN BUDGET

SELECTION NO. 02

Oud Wood

Tom Ford

Oud | Sandalwood | Rosewood | Amber | 87%

"Formal evening silhouette — resinous warmth for smart casual."

$195.00 Nordstrom · WITHIN BUDGET

SELECTION NO. 03

Baccarat Rouge 540

Maison F. Kurkdjian

Jasmine | Saffron | Amberwood | Fir | 81%

"Luminous crystalline amber — commands every room."

$340.00 Bloomingdale's · WITHIN BUDGET

Summary & Future Directions

WHAT WE BUILT

Pipeline Summary

- ✓ LLM enriched dataset of 36,000 fragrances pulled across 5 datasets
- ✓ Zero-shot CLIP classifier with prompt-engineered vibe embeddings
- ✓ Sentence Transformer for semantic description matching
- ✓ 5-dimensional vibe representation of user inputs and fragrance data
- ✓ Recommendations using cosine similarity and optimized weights
- ✓ Real-time price discovery via SerpAPI Google Shopping engine
- ✓ Top 3 matching fragrances to match the user's outfit and occasion

FUTURE DIRECTIONS

Model Evolution

Dataset	Filter fragrance database for popular fragrances
Enrichment	Analyze fragrances individually for high accuracy
CLIP	Fine-tune CLIP on outfit data and associated vibes
Qwen3	Fine-tune on fragrance language and context
Vibe	Increase vibe dimensions to represent fragrances better
Recommend	Highly dependent on models used to generate vibes
Webscrape	Only recommend and search for popular fragrances

Dress for the occasion. Scent it perfectly.